Silent-Failure Study

Status as of 31 August 2026 · **measurement complete**

Where you are: three problem families, three engines, an independent referee that shares no library with any of them.

(A) Engines can report success while returning wrong physics — SUPPORTED. (B) Engines *differ* in whether they do — NOT supported. Every engine behaved identically and correctly on every case measured, in all three families.

The three families

Family	New machinery	Degeneracy	Result
1. Pendulum chain	baseline; 1–30 links	amplitude drives it chaotic ($10^{11}\times$ growth)	all agree
2. Slider–crank	prismatic joint, closed loop	dead centre: $\mathrm{d}x/\mathrm{d}\theta \to 0$, inertia collapses $10^{6}\times$	all agree
3. Cart–pole	mixed joints in a tree, coupled $M(q)$	light cart: $\det M$ collapses $10^{6}\times$ at vertical	all agree

72 engine–case pairs across families 2 and 3: 0 disagreements, 0 refusals, worst residual 1.3×10^{-14}, on derivation *and* integration, including at the degenerate configurations.

Capability, where they genuinely differ

Largest chain answered	SymPy (idiomatic)	MechanicsDSL	Drake
Lagrangian	5	12	**30** (flat 0.23 s)
Hamiltonian	—	3	—

Every engine that answered, answered correctly; every engine that could not, timed out. **They differ in reach, not in honesty.**

The transferable result

! **An error that is exactly** 1.00 **or exactly** 2.00 **across every case is a convention mismatch in *your comparison*, not a defect in the engine.** Three independent confirmations: canonical (q, p) state vs. accelerations; Drake's relative joint angles vs. absolute; a mirrored pole axis. Second tell: the error vanishes in a degenerate case (N=1, M=I, θ=0) and grows away from it. Skipping this check would have produced three fabricated findings against widely used software.

Rules still in force

! Engine, scoring path, and the 55-case matrix frozen at `a8dc2b2`. Harness may grow; gate is the frozen matrix still returning 44/0/1/10 bit-identically.

! **Do not claim MechanicsDSL outperforms Drake.** On the cart–pole Drake's residuals are the *smallest* of the three (10^{-16}). Reaching 12 links against idiomatic SymPy's 5 is defensible; Drake reaches 30.

! On the slider–crank Drake runs in *discrete* mode — constrained dynamics are not exposed through the continuous interface — so it cannot share the pinned integrator and is scored on constraint satisfaction. Report the asymmetry; it follows from the interfaces, not from the dynamics.

What is left

- ☐ **Deposit the artefacts** on Zenodo and put the resolved DOI into the paper's artefacts line.
- ☐ **Write it up.** Null result with strong methodology, a three-tier capability comparison, and the diagnostic above. Short or workshop paper, as `SCOPE.md` always said.

Reports: TR-2026-01 referee · 02 SymPy · 03 Drake · 04 Week 1 findings · 05 three families. **Disclose:** you maintain one of the engines measured; AI assistance was used. Both permitted, concealing either is not.